

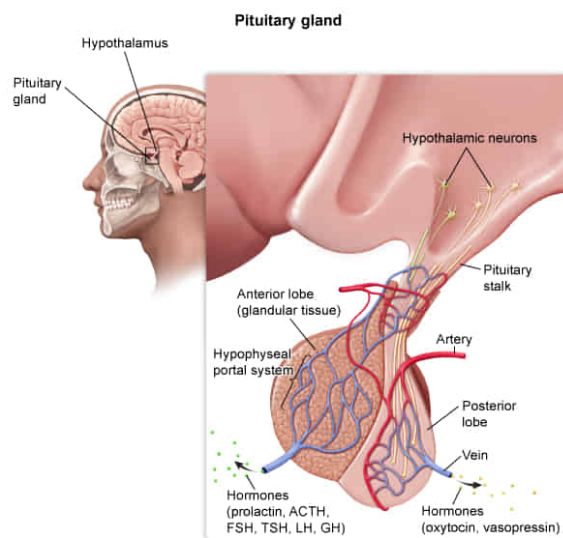
Which statement best describes the characteristics of hormones secreted from the pituitary gland?

- ✔ ☒ A. Hormones secreted from the posterior pituitary are synthesized in the hypothalamus. [66%]
- ☐ B. Hormones secreted from the posterior pituitary act on other endocrine glands. [21%]
- ☐ C. Hormones secreted from the anterior pituitary act directly only on nonendocrine tissues. [4%]
- ☐ D. Hormones secreted from the anterior pituitary are released by depolarization of the nerve terminals. [7%]

Correct

65% answered correctly

Explanation:



The **pituitary gland** is a small structure that sits inside a bony cavity below the hypothalamus at the base of the brain. It is attached to the hypothalamus via the pituitary stalk and can be divided into two separate lobes:

- The **anterior** lobe is made up of **glandular tissue** containing different cell types that synthesize and secrete follicle-stimulating hormone (FSH), luteinizing hormone (LH), adrenocorticotrophic hormone (ACTH), thyroid-stimulating hormone (TSH), growth hormone (GH), β -endorphins, and prolactin. The synthesis and secretion of hormones from the **anterior pituitary** is controlled primarily by neurohormones released from hypothalamic neurons. These regulatory neurohormones are secreted into the **hypophyseal portal system**, a venous network of small blood vessels that enables small quantities of hormones secreted from the hypothalamus to directly reach the anterior pituitary without being diluted in the circulation.
- The **posterior** lobe, made up of **axonal projections** from the hypothalamus, stores

and secretes the hormones oxytocin and vasopressin (also called antidiuretic hormone [ADH]). **Posterior pituitary hormones** are synthesized in hypothalamic neurons and then transported down the axon to the axon terminals in the posterior pituitary, where they are stored. The secretion of stored hormones from the posterior pituitary is mediated by action potentials that cause exocytosis of neurosecretory vesicles.

(Choices B and C) Hormones can be classified by their modes of actions. Tropic hormones regulate the secretion of other hormones by acting on other endocrine glands. In contrast, direct or nontropic hormones act directly on nonendocrine tissues to cause physiological end points (eg, ADH-induced **water reabsorption**). The anterior pituitary releases both tropic (FSH, LH, ACTH, TSH, GH) and direct (GH, prolactin, endorphins) hormones, whereas the posterior pituitary releases only direct hormones (oxytocin, ADH).

(Choice D) The secretion of hormones from the anterior pituitary is stimulated primarily by regulatory neurohormones released into the blood by hypothalamic neurons. In contrast, the secretion of stored hormones from the posterior pituitary is induced by depolarization of the nerve terminals.

Educational objective:

The synthesis and secretion of hormones from the anterior pituitary is regulated by the release of neurohormones into the blood from neurons located in the hypothalamus. In contrast, posterior pituitary hormones are synthesized in hypothalamic neurons and undergo anterograde axonal transport to the posterior pituitary. The secretion of stored posterior pituitary hormones from the axon terminals is mediated by depolarization of the nerve terminals.

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Subject:

Biology

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System:

Endocrine and Nervous Systems